

Max Bielstein

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(402) 213-8842

EXPERIENCE

Principal Financial Group

SOFTWARE ENGINEER I - II

June 2024-Current | Des Moines, IA (Remote)

- Created a high volume internal API using AWS Lambda and DynamoDB to store and deliver customer-facing data
- Managed team application operations during on-call shifts, actively monitoring metrics and escalating issues when needed.
- Built GitHub Actions workflows to automate the disaster recovery processes for our applications
- Designed and implemented a scalable sync solution using Typescript to keep data up to date using parallel asynchronous calls between several different internal APIs
- Used Domain Driven Design to build up domain APIs using an Event Messaging architecture

SOFTWARE ENGINEERING INTERN

May 2023-June 2024 | Des Moines, IA (Remote)

- Upgraded and maintained multiple Amazon Web Services (AWS) applications that incorporate domain-driven design
- Created and maintained GitHub Actions workflows to automate the testing and deployment of applications
- Implemented AWS alarms to track errors and alert the team when they occur

Thawe LLC

Cofounder

August 2025-Current

- Designed and implemented a professional networking mobile app to allow users to connect with like minded professionals in their area
- Architected a Django backend prioritizing security and release velocity
- Prioritized and implemented features based on market data and customer feedback

Kiewit

SOFTWARE ENGINEERING INTERN

May 2022-August 2022 | Omaha, NE (Hybrid)

- Developed features to corporate applications while using scrum and agile development methodologies
- Worked with other interns on a scrum team to develop features on an existing application
- Added API calls and other functionality to a .NET application

University of Nebraska-Lincoln

SOFTWARE ENGINEERING TEACHING ASSISTANT

August 2021-May 2023 | Lincoln, NE

- Helped students in understanding and completing their coursework
- Assisted in directing and grading labs
- Explained difficult to understand concepts to first-time programmers

EDUCATION

UNIVERSITY OF NEBRASKA-LINCOLN

BACHELOR OF SCIENCE

SOFTWARE ENGINEERING

GPA: 3.95

August 2020 - May 2024

LINKS

LinkedIn: www.linkedin.com/in/maxbielstein

Website: www.maxbielstein.com

COURSEWORK

Database Design

Algorithms

UX/UI

Formal Verification

Machine Learning AI

Software Architecture

Testing and Analysis

Software Engineering

Agile Processes

SKILLS

LANGUAGES

Python

C#

Typescript

CSS

SQL

C++

Java

Javascript

HTML

Dafny

C

Perl

Tools and Frameworks

Git

REST APIs

Azure DevOps

AWS

Jira

Yarn

Node.js

Spring

Next.js

DynamoDB

React

.NET

Agile

GitHub Actions

Docker

PNPM

TensorFlow

Sinon

Chai

Google Cloud

PROJECTS

Thawe | Thawe is a professional networking app that connects entrepreneurs and professionals through a swipe-based matching system, enabling meaningful business relationships and collaboration opportunities.

- Built a cross-platform professional networking mobile app (iOS & Android) using React Native, Expo, and TypeScript, featuring a swipe-based discovery interface, real-time messaging, and location-based search integrated with Firebase and a custom REST API
- Applied agentic AI to add documentation, fix bugs, and develop features within the app using Github Copilot, Claude Code, and Codex
- Architected a hybrid data model combining PostgreSQL for structured profile data with Firestore for real-time messaging, integrating Firebase Cloud Messaging for push notifications and Sightengine AI for automated content moderation at upload time.
- Engineered a location-based professional networking backend using Django REST Framework, PostgreSQL/PostGIS, and Firebase Auth, enabling geospatial user discovery with configurable radius filtering and full-text profile search with weighted relevance ranking.

Next.js AWS Website Template | Template website for quickly creating and deploying websites out to AWS

- Engineered a multi-region (active/passive) disaster recovery architecture on AWS with Route53 health checks and automatic DNS failover, achieving cross-region resilience with no client-side changes.
- Delivered a serverless Next.js 15 app on AWS Lambda via the Lambda Web Adapter, fronted by API Gateway and CloudFront for low-cost, auto-scaling hosting.
- Authored a reusable AWS CDK (TypeScript) infrastructure template provisioning Lambda, CloudFront, Route53, ACM, and WAF from a single configurable interface.

AWS Serverless API Template | Template website for quickly creating and deploying AWS Lambda APIs

- Architected a multi-region active/passive AWS API with automated Route53 health-check DNS failover for disaster recovery.
- Built a reusable serverless API template as TypeScript AWS CDK infrastructure-as-code, with isolated dev/staging/prod environments.
- Deployed API Gateway v2 + Lambda behind a global CloudFront edge with ACM TLS and custom domains.

Pizza Visualization Application | Application for displaying pizzas within a pizza oven without the use of interior cameras

- Designed and created the user interface with JavaFX
- Coded the algorithm in Java for displaying pizzas within an oven with the use of exterior cameras
- Used webcam APIs to access, manipulate, and display images

Relvo Alloy | An application that uses machine learning to speed up the process of solving genetic algorithms in formal verification processes

- Created a reinforcement learning algorithm to optimize the speed of an existing formal verification software called [EvoAlloy](#)
- Successfully decreased the time needed for the formal verification to complete in many cases